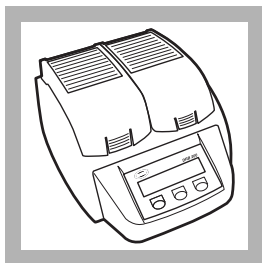


NITROGEN, TOTAL, Test 'N Tube (0.0 to 25.0 mg/L N)

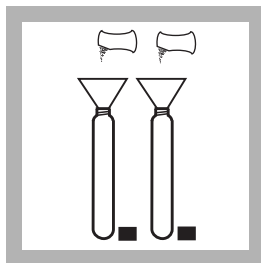
TNT Persulfate Digestion Method

For water and wastewater



1. Turn on the DRB 200 Reactor. Heat to 103-106 °C (optimum temperature is 105 °C).

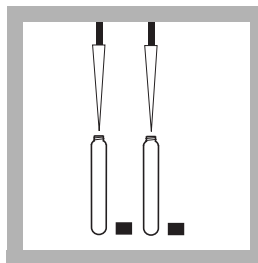
Note: For proof of accuracy, run a 20 mg/L $\text{NH}_3\text{-N}$ standard through digestion and analysis.



2. Using a funnel, add the contents of one Total Nitrogen Persulfate Reagent Powder Pillow to each of two Total Nitrogen Hydroxide Reagent vials.

Note: Wipe off any reagent that may get on the lid or the tube threads.

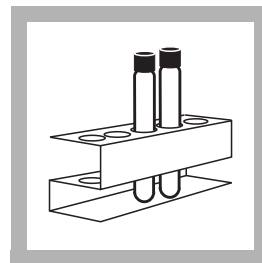
Note: One reagent blank is sufficient for each set of samples.



3. Add 2 mL of sample to one vial. Add 2 mL of organic-free water to another vial (the reagent blank). Cap both vials and shake vigorously (about 30 seconds). Place the vials in the Reactor. Heat for 30 minutes.

Note: The reagent may not dissolve completely after shaking.

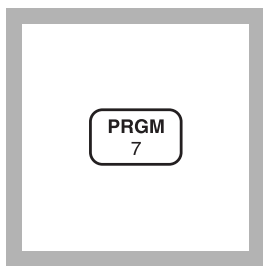
Note: Alternate water must be free of all nitrogen-containing species.



4. Using finger cots or gloves, remove the hot vials from the reactor and allow to cool to room temperature.

Note: It is very important to remove the vials from the Reactor after exactly 30 minutes.

NITROGEN, TOTAL, Test 'N Tube, continued

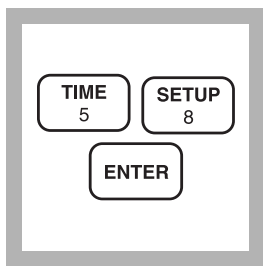


5. Enter the stored program number for Test 'N Tube Total Nitrogen.

Press: **PRGM**

The display will show:

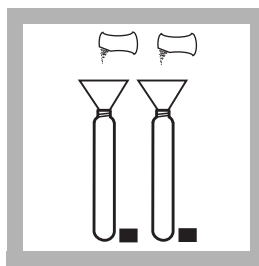
PRGM ?



6. Press: **58 ENTER**

The display will show **mg/L, N** and the **ZERO** icon.

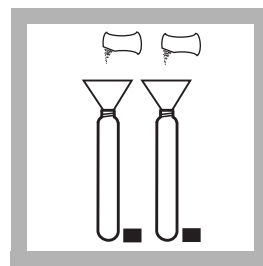
*Note: For alternate forms (NH_3 , NO_3), press the **CONC** key.*



7. Remove the caps from the digested vials and add the contents of one TN Reagent A Powder Pillow to each vial. Cap the vials and shake for 15 seconds.

Press: **TIMER ENTER** after shaking.

A three-minute reaction period will begin.



8. After the timer beeps, remove the caps from the vials and add one TN Reagent B Powder Pillow to each vial. Cap the vials and shake for 15 seconds. The display will show:

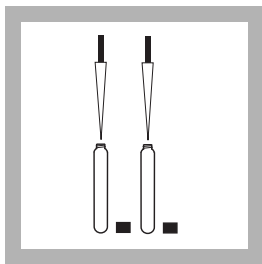
02:00 Timer 2

Press **ENTER** after shaking.

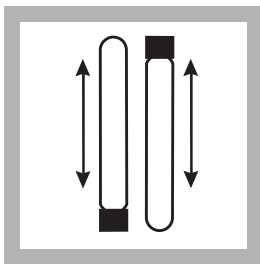
A two-minute reaction period will begin.

Note: The reagent will not completely dissolve. The solution will begin to turn yellow.

NITROGEN, TOTAL, Test 'N Tube, continued

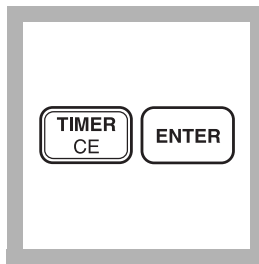


9. After the timer beeps, remove the caps from two TN Reagent C Vials. Add 2 mL of digested, treated sample to one vial. Add 2 mL of the digested, treated reagent blank to the second TN Reagent C Vial.



10. Cap and invert 10 times to mix. Use slow, deliberate inversions for complete recovery. The vials will be warm.

Note: Follow these instructions for inversion or low results may occur. Hold the vial vertical with the cap up. Invert the vial and wait for all of the solution to flow to the cap end. Pause. Return the vial to the upright position and wait for all of the solution to flow to the vial bottom. This is one inversion (10 inversions = 30 seconds).

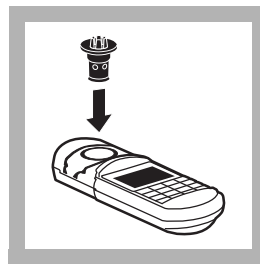


11. The display will show: **05:00 Timer 3**

Press: **ENTER**

A five-minute reaction period will begin.

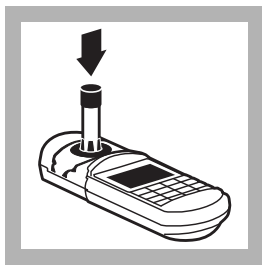
Note: The yellow color will intensify.



12. During the reaction period, insert the COD/TNT Adapter into the cell holder by rotating the adapter until it drops into place. Then push down to fully insert it.

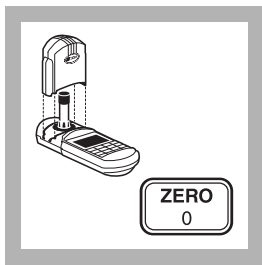
Note: For increased performance, a diffuser band covers the light path holes on the adapter. Do not remove the diffuser band.

NITROGEN, TOTAL, Test 'N Tube, continued



13. After the timer beeps, wipe the TN Reagent C vial containing the reagent blank. Place the vial in the adapter. Push straight down on the top of the vial until it seats solidly into the adapter.

Note: Do not move the vial from side to side as this can cause errors.



14. Tightly cover the vial with the instrument cap.

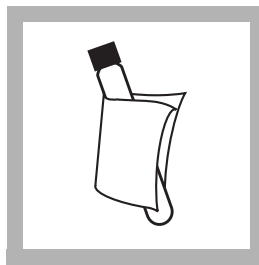
Press: **ZERO**

The cursor will move to the right, then the display will show:

0.0 mg/L N

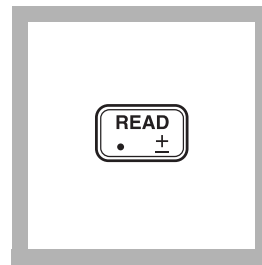
Note: Wiping with a damp towel, followed by a dry one, will remove fingerprints or other marks.

Note: The reagent blank is stable when stored in the dark; see Blanks For Colorimetric Measurement following these steps.



15. Wipe the TN Reagent C vial containing the sample and place it into the adapter. Tightly cover the vial with the instrument cap.

Note: Multiple samples may be read after zeroing on one reagent blank.



16. Press: **READ**

The cursor will move to the right, then the result in mg/L nitrogen (N) will be displayed.

Note: Standard Adjust may be performed using a prepared standard (see Standard Adjust in Section 1).

Note: If the display flashes "limit", dilute the sample and repeat the digestion and the colorimetric finish. The digestion must be repeated for accurate results; diluting and repeating the color finish does not yield complete results. Multiply the result by the dilution factor; see Section 1.

Sampling and Storage

Collect samples in clean plastic or glass bottles. Best results are obtained with immediate analysis.

Preserve the sample by reducing the pH to 2 or less with concentrated sulfuric acid (at least 2 mL). Store at 4 °C (39 °F) or less. Preserved samples may be stored up to 28 days. Warm samples to room temperature and neutralize with 5 N sodium hydroxide before analysis. Correct the test result for volume additions; see *Correcting for Volume Additions* in Section 1.

Accuracy Check

This method generally yields 95-100% recovery on organic nitrogen standards. For proof of accuracy Hach offers a set of three Primary Standards for Kjeldahl Nitrogen.

NITROGEN, TOTAL, Test 'N Tube, continued

1. Prepare one or more of the following three solutions. Each preparation is for an equivalent 25 mg/L N standard. Use water that is free of all organic and nitrogen-containing species.
 - a) Weigh 0.3379 g of Ammonium p-Toluenesulfonate (PTSA). Dissolve in a 1000-mL volumetric flask with deionized water. Add deionized water to the 1000-mL mark.
 - b) Weigh 0.4416 g of Glycine p-Toluenesulfonate. Dissolve in a 1000-mL volumetric flask with deionized water. Add deionized water to the 1000-mL mark.
 - c) Weigh 0.5274 g of Nicotinic p-Toluenesulfonate. Dissolve in a 1000-mL volumetric flask with deionized water. Add deionized water to the 1000-mL mark.
2. Analyze each of these solutions using the test procedure above. Calculate the percent recovery for each using this formula:

$$\% \text{ recovery} = \frac{\text{measured concentration}}{25} \times 100$$

The percent recovery should be:

Compound	Lowest Expected % Recovery
Ammonia-PTSA	95%
Glycine-PTSA	95%
Nicotinic-PTSA	95%

Hach analysts have found Ammonia-PTSA to be the most difficult to digest. Other compounds may yield different percent recoveries.

Standard Solution Method

Substitute 2 mL of a 20 mg/L ammonia nitrogen standard solution for the sample. To prepare a 20-mg/L standard, use a 20-mL Class A pipet to transfer 20 mL of a 100-mg/L Ammonia Nitrogen Standard (see *Optional Reagents*) to a 100-mL Class A volumetric flask. Dilute to the line with organic-free water. A single analyst should obtain less than 5% variation on replicates. Comparison of the user-obtained value with the standard concentration is an indication of test performance for this user.

NITROGEN, TOTAL, Test 'N Tube, continued

Standard Additions Method

- a) Fill three 25-mL graduated mixing cylinders with 25 mL of sample.
- b) Snap the neck off an Ammonia Nitrogen Voluette Ampule Standard Solution, 160 mg/L as $\text{NH}_3\text{-N}$.
- c) Use the TenSette Pipet to add 0.3 mL, 0.6 mL, and 0.9 mL of standard, respectively, to the three mixing cylinders.
- d) Stopper each cylinder and mix thoroughly.
- e) Add 2 mL of each prepared solution, respectively, to three TN Hydroxide Reagent Sample Digestion Vials.
- f) Analyze each standard addition sample as described in the procedure. The nitrogen concentration should increase 2 mg/L for each 0.3 mL of standard added.
- g) If these increases do not occur, see *Standard Additions* in *Section 1* for troubleshooting information.

Blanks for Colorimetric Measurement

The reagent blank may be used up to 7 days for measurements using the same lots of reagents. Store the reagent blank in the dark at room temperature (18-25 °C). If a small amount of white floc appears prior to the end of one week, discard the reagent blank and prepare a new one.

Method Performance

Precision

A Hach chemist analyzed two independent nutrient standards. The lowest average percent recovery was 95% with a standard deviation of $\pm 2\%$.

In a single laboratory, using a standard solution of 15.0 mg/L N and two lots of reagent with the instrument, a single operator obtained a standard deviation of less than ± 0.5 mg/L N. For more information on Hach's precision statement, see *Section 1*.

Estimated Detection Limit

The estimated detection limit for program 58 is 2 mg/L N. For more information on the estimated detection limit, see *Section 1*.

NITROGEN, TOTAL, Test 'N Tube, continued

Interferences

Interfering substances that resulted in a concentration change of $\pm 10\%$:

Substance	Level and Effect
Bromide	>60 ppm; positive interference
Chloride	>1000 ppm; positive interference

The substances in the following table have been tested and found **not** to interfere up to the indicated levels (in mg/L):

Substance	Maximum Level Tested (mg/L)
Barium	2.6
Calcium	300
Chromium (3+)	0.5
Iron	2
Lead	6.6 ppb
Magnesium	500
Organic Carbon	150
pH	13 pH units
Phosphorus	100
Silica	150
Silver	0.9
Tin	1.5

Hach chemists tested this chemistry on standard nitrogen solutions prepared from the following compounds and obtained $\geq 95\%$ recovery:

- Ammonium chloride
- Ammonium sulfate
- Ammonium acetate
- Urea
- Glycine

Ammonium chloride or nicotinic-PTSA spikes in domestic influent, effluent and the ASTM standard specification for substitute wastewater (D 5905-96) also resulted in $\geq 95\%$ recovery.

NITROGEN, TOTAL, Test 'N Tube, continued

Large amounts of nitrogen-free organic compounds in some samples may decrease digestion efficiency by consuming some of the persulfate reagent. Samples known to contain high levels of organics should be diluted and re-run to verify digestion efficiency.

Summary of Method

An alkaline persulfate digestion converts all forms of nitrogen to nitrate. Sodium metabisulfite is added after the digestion to eliminate halogen oxide interferences. Nitrate then reacts with chromotropic acid under strongly acidic conditions to form a yellow complex with an absorbance maximum near 420 nm.

REQUIRED REAGENTS

Description	Cat. No.
Test 'N Tube Total Nitrogen Reagent Set (50 vials).....	26722-45
Includes:	
TN Reagent C Vials, Acid Solution*, 50/pkg	26721-45
TN Hydroxide Reagent Sample Digestion Vials*, 50/pkg.....	26717-45

Description	Quantity Required		Unit	Cat. No.
	Per Test			
TN Persulfate Reagent Powder Pillows.....	2 pillows	100/pkg		26718-49
TN Reagent A, Bisulfite Powder Pillows.....	2 pillows	100/pkg		26719-49
TN Reagent B, Indicator Powder Pillows.....	2 pillows	100/pkg		26720-49

REQUIRED APPARATUS

DRB 200 Reactor, 110 V, 15 x 16 mm tubes	LTV082.53.40001		
DRB 200 Reactor, 220 V, 15 x 16 mm tubes	LTV082.52.40001		
COD/TNT Adapter	1	each	48464-00
Funnel, micro	1	each	25843-35
Pipet, TenSette, 1.0-10.0 mL	1	each	19700-10
Pipet Tips for 19700-10.....	1	50/pkg	21997-96
Pipet, TenSette, 0.1 to 1.0 mL	1	each	19700-01
Pipet Tips for 19700-01.....	2	50/pkg	21856-96
Test Tube Cooling Rack.....	1-3	each	18641-00

* Not available separately.

NITROGEN, TOTAL, Test 'N Tube, continued

OPTIONAL REAGENTS

Description	Quantity Required		Unit	Cat. No.
	Per Test			
Nitrogen, Ammonia, 100 mg/L $\text{NH}_3\text{-N}$	500 mL		24065-49	
Nitrogen, Ammonia, Voluette Ampule, 160 mg/L $\text{NH}_3\text{-N}$, 10 mL	16/pkg		21091-10	
Sulfuric Acid, ACS	500 mL		979-49	
Primary Standards for Kjeldahl Nitrogen.....	set of 3		22778-00	
Ammonium p-Toluenesulfonate.....	25 g		22779-24	
Glycine p-Toluenesulfonate	25 g		22780-24	
Nicotinic Acid p-Toluenesulfonate	25 g		22781-24	
Sodium Hydroxide Standard Solution, 5.0 N.....	50 mL	MDB	2450-26	
Wastewater Effluent Standard, Inorganics ($\text{NH}_3\text{-N}$, $\text{NO}_3\text{-N}$, PO_4 , COD, SO_4 , TOC).....	500 mL		28332-49	
Water, organic-free	500 mL		26415-49	

OPTIONAL APPARATUS

Ampule Breaker Kit	each		21968-00	
Balance, analytical, 115 VAC.....	each		28014-01	
Balance, analytical, 230 VAC	each		28014-02	
Cots, finger	2/pkg		14647-02	
Cylinder, graduated, mixing, 25 mL (3 required)	each		26363-40	
Flask, volumetric, Class A, 1000 mL (3 required).....	each		14574-53	
Flask, volumetric, Class A, 100 mL.....	each		14574-42	
Pipet, volumetric, Class A, 20 mL	each		14515-20	
Pipet Tips, for 19700-01 TenSette Pipet	1000/pkg		21856-28	
pH Paper, 1 to 11 pH units	5 rolls/pkg		391-33	
DRB 200 Reactor, 110 V, 21 x 16 mm and 4 x 20 mm	LTV082.53.42001			
DRB 200 Reactor, 220 V, 21 x 16 mm and 4 x 20 mm.....	LTV082.52.42001			
DRB 200 Reactor, 110 V, 9 x 16 mm and 2 x 20 mm.....	LTV082.53.30001			
DRB 200 Reactor, 220 V, 9 x 16 mm and 2 x 20 mm.....	LTV082.52.30001			